Conditional State Edit Field for Low-Light Image Enhancement

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Abstract

Low-light enhancement aims to recover visibility, color consistency, and structural detail from underexposed images. Existing methods have made substantial progress through Retinex decomposition, curve estimation, unsupervised learning, and Transformer-based modeling, but most of them learn a direct mapping from low-light images to normally exposed images and provide limited explicit editing semantics or inference-time controllability. This paper proposes Conditional State Edit Field (CSEF), which formulates low-light enhancement as a state-conditioned atomic editing process. CSEF constructs an eight-dimensional pixel state space, generates image-conditioned editing atoms with a lightweight controller, matches pixels to atoms through soft assignments in the state space, and synthesizes a dense per-pixel edit field. A differentiable renderer then converts the edit field into deterministic operations for luminance, chroma, white balance, detail, denoising, and shadow lifting. Experiments on LOL-v2 Synthetic show that CSEF achieves the best PSNR, SSIM, and MAE among the compared methods, while on LOL-v2 Real it provides a stable quality–efficiency trade-off with about 124K parameters and high inference speed. Further experiments demonstrate inference-time controllability and transferability to underwater and color enhancement tasks.

1. Introduction

Low-light enhancement aims to recover visibility, color consistency, and structural details from underexposed images. Existing methods have achieved substantial progress along several directions, including Retinex decomposition, curve estimation, unsupervised learning, and Transformer-based modeling. However, most approaches mainly learn a direct mapping from low-light images to normally exposed images, and therefore lack explicit editing semantics and inference-time controllability.

This paper proposes Conditional State Edit Field (CSEF), which formulates low-light enhancement as a state-conditioned atomic editing process. Specifically, CSEF first constructs an eight-dimensional pixel state space using fixed image statistics. A lightweight controller then generates 16 image-conditioned editing atoms, each of which contains a state-space center and the magnitudes of predefined edits. Each pixel is softly matched to the atoms according to the similarity between its state and the atom centers, and the matched atoms are combined into a per-pixel edit field. Finally, a differentiable renderer converts the edit field into deterministic operations, including luminance adjustment, chroma control, white balance, detail modulation, denoising, and shadow lifting.

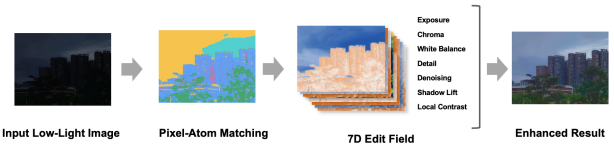


Figure 1. Conceptual illustration of CSEF. Pixel–atom matching produces a seven-dimensional edit field that controls interpretable enhancement operations.

CSEF establishes an interpretable path from pixel states to atom matching, editing parameters, image operations, and enhanced results. On LOL-v2 Synthetic, CSEF achieves the best PSNR, SSIM, and MAE among the compared methods. On LOL-v2 Real, the model provides a stable quality–efficiency trade-off with about 124K parameters and high FPS. Further experiments show that CSEF supports inference-time adjustment and can be transferred to underwater enhancement and color enhancement tasks, suggesting that the proposed atomic editing mechanism is extensible.

The main contributions are summarized as follows:

- We propose CSEF, a state-conditioned atomic editing framework for interpretable low-light enhancement.
- We design an eight-dimensional imaging state space and a pixel–atom matching mechanism to realize spatially adaptive editing.
- We validate the effectiveness of CSEF through low-light enhancement, ablation, controllability, and transfer experiments.

2. Related Work

2.1. Low-Light Enhancement

Low-light enhancement aims to recover visibility, color, and structural details from underexposed images while suppressing the noise amplification commonly observed in dark imaging. Retinex theory is an important modeling foundation in this field. RetinexNet [1] decomposes a low-light image into reflectance and illumination components and learns illumination enhancement and reflectance restoration separately. KinD [2] further organizes enhancement into image decomposition, illumination adjustment, and reflectance restoration, giving brightness correction, color recovery, and noise suppression clearer functional roles. For unsupervised learning, EnlightenGAN [3] learns low-light enhancement from unpaired data, Zero-DCE [4] formulates enhancement as image-adaptive curve estimation, and SCI [5] achieves efficient enhancement through self-calibrated illumination learning. These methods advance low-light enhancement from the perspectives of Retinex decomposition, unpaired learning, zero-reference curve estimation, and illumination calibration.

Recent methods further emphasize complex degradation, spatially adaptive restoration, and noise-aware modeling. SNR-Aware Transformer [6] uses signal-to-noise-ratio distributions to guide different enhancement strategies for different regions, reducing noise amplification in dark areas. LLFlow [7] uses normalizing flows to model the one-to-many mapping from low-light images to normally exposed images, improving the expressive flexibility of enhancement results. Retinexformer [8] combines a Retinex-inspired one-stage framework with an illumination-guided Transformer to strengthen long-range interactions across illumination regions. These methods show that low-light enhancement must jointly handle the coupling among brightness, color, detail, and noise, but they do not explicitly expose these operations as interpretable editing processes. Following the direction of joint modeling, this paper represents low-light enhancement as a predefined editing process with clear operational semantics in a state space. Through an explicit edit field, CSEF performs brightness, color, white balance, detail, denoising, and shadow-lifting operations.

2.2. Matching-Based Representations

Using a finite number of prototypes, codewords, or atoms to organize complex data is a classical idea in machine learning and visual representation. Sparse coding [10] represents natural image structures with atoms from an overcomplete dictionary. VQ-VAE [11] learns a discrete codebook and maps continuous latent features to codebook vectors, forming discrete representations through code assignment.

Similar primitive-combination ideas also appear in image enhancement and image editing. HDRNet [12] predicts local affine color transforms in a bilateral grid and selects transforms according to each pixel's spatial and color coordinates. DeepLPF [13] learns spatially varying local parametric filters, making enhancement correspond to interpretable local editing operations. Image-adaptive 3D LUT [14] learns multiple basis LUTs and predicts image-conditioned fusion weights, while spatial-aware 3D LUT [15] introduces pixel-level LUT fusion maps so that different regions can use different combinations of editing bases. These methods indicate that image enhancement can be achieved with reusable editing primitives and content-related matching or fusion. This paper instantiates this idea as an atomic editing framework for low-light enhancement: CSEF constructs an eight-dimensional state space, initializes atom centers and scales with data-aware k-means, and produces per-pixel edit fields through pixel–atom softmax matching.

3. Method

3.1. Model Overview

CSEF is a low-light image enhancement model based on a Conditional State Edit Field. As illustrated in Fig. 2, it represents image enhancement as an editing program generated by a neural network and executed by deterministic image-processing operators. The program consists of a finite set of editing atoms, pixel–atom matching relationships, and a spatial denoising prior.

Given a low-light image $I \in [0,1]^{3\times H\times W}$, the model first constructs a pixel state representation with explicit imaging meanings. A lightweight controller generates $K = 16$ editing atoms according to the state distribution of the whole image. Each atom describes a class of pixel states and the intensity values of a series of classical enhancement operations, including luminance adjustment, chroma control, white balance correction, detail modulation, noise suppression, and shadow lifting. Each pixel obtains soft matching weights according to the similarity distance between its state and the atom centers, and then synthesizes a continuous per-pixel edit field. Finally, the renderer converts the edit field into concrete image operations and outputs the enhanced result.

The overall computation is

$$\begin{aligned} S &= \Phi(I), \\ (A, Q) &= C_\theta(S), \\ E &= \mathcal{M}(S, A), \\ \widehat{I} &= \mathcal{R}(I, E, Q), \end{aligned} \tag{1}$$

where Φ is the state feature construction function, C_θ is the image-level controller, A is the set of editing atoms, Q is the spatial denoising prior, $\mathcal{M}$ is the atom matching and edit-field synthesis function, and $\mathcal{R}$ is the differentiable image

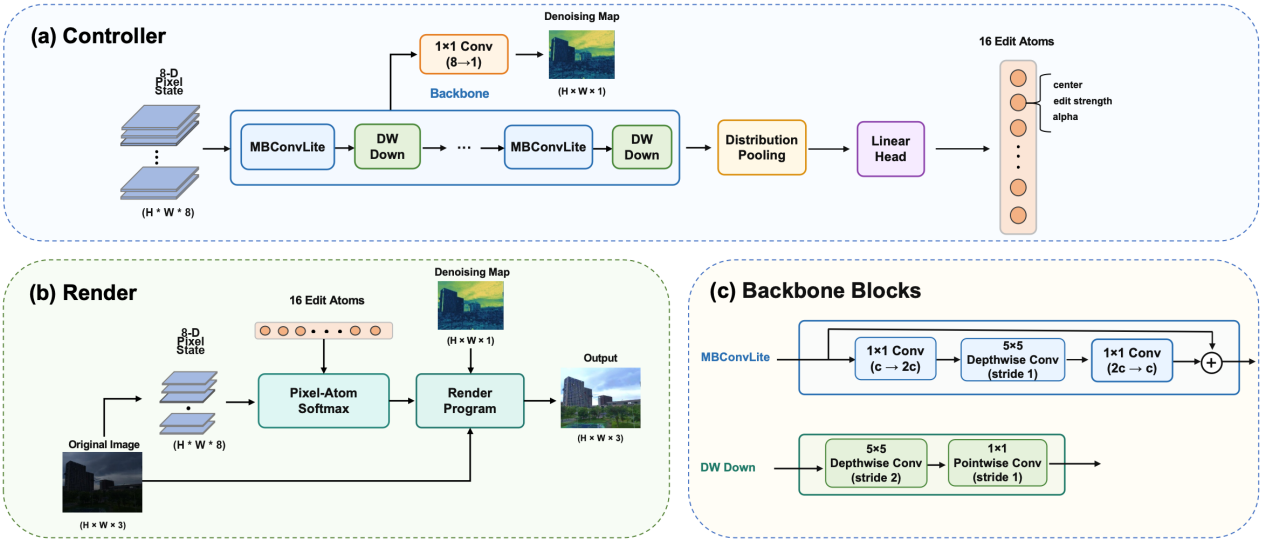


Figure 2. Overall architecture of CSEF. The controller takes the enhancement state as input and predicts a denoising map and 16 edit atoms through lightweight backbone blocks and distribution pooling. The renderer performs pixel–atom softmax matching and executes the render program to produce the enhanced image.

renderer. This structure forms an interpretable path:

$$\begin{aligned}\text{pixel state} &\rightarrow \text{atom matching} \rightarrow \text{edit parameters} \\ &\rightarrow \text{image operations} \rightarrow \text{enhanced result}.\end{aligned} \tag{2}$$

Any output pixel can be traced back to the weighted contributions of a finite number of atoms and further associated with concrete brightness, color, detail, and denoising operations.

3.2. Image State Representation

The input image is first converted into an eight-dimensional state tensor:

$$S = \Phi(I) \in [0,1]^{8\times H\times W}. \tag{3}$$

For pixel location p, the state vector is defined as

$$s_p = \begin{bmatrix} l_p^{rel} & l_p & u_p & v_p & g_p & c_p & t_p & n_p \end{bmatrix}^T, \tag{4}$$

where l_p^{rel} is the log-luminance relative to the average exposure level of the whole image, l_p is normalized absolute log-luminance, u_p and v_p are normalized YUV chroma components, g_p is the local luminance-normalized gradient, c_p is local contrast, t_p is texture strength, and n_p is a dark-region noise proxy. This state space directly corresponds to the main decision factors in low-light enhancement, allowing pixels with similar imaging states to share editing strategies.

3.3. Image-Conditioned Editing Atoms

The controller C_θ extracts image-level context from the whole state map and generates the editing atoms for the current input image. It uses a lightweight convolutional encoder:

$$\begin{aligned}8 &\xrightarrow{\text{Conv}_{3\times3,s=2}} 24 \xrightarrow{\text{MBConvLite}} 24 \xrightarrow{\text{DWDown}} 48 \\ &\xrightarrow{\text{MBConvLite}} 48 \xrightarrow{\text{DWDown}} 80 \xrightarrow{\text{MBConvLite}} 80 \xrightarrow{\text{Conv}_{1\times1}} 64.\end{aligned} \tag{5}$$

MBConvLite consists of a 1×1 channel expansion, a 5×5 depthwise convolution, a 1×1 channel projection, and a residual connection. DWDown performs downsampling with depthwise and pointwise convolutions.

The encoded features are converted into image-level context by distribution pooling. For each channel, the model computes the mean, standard deviation, soft high response, and soft low response:

$$z = \begin{bmatrix} \text{mean}(F) & \text{std}(F) & \text{softHigh}(F) & \text{softLow}(F) \end{bmatrix}. \tag{6}$$

These statistics describe the global level, dispersion, and high- and low-response regions of the feature map. The concatenated vector is passed through a two-layer MLP to generate $K = 16$ editing atoms. The i-th atom is

$$a_i = (c_i, e_i, \alpha_i), \tag{7}$$

where $c_i \in [0,1]^8$ is the state center, $e_i \in \mathbb{R}^7$ is the edit vector, and $\alpha_i \in [0,1]$ is the atom strength. The seven-dimensional edit vector is

$$e_i = \begin{bmatrix} \Delta l_i & \gamma_i^c & b_i^{rg} & b_i^{bg} & \gamma_i^d & \eta_i & s_i \end{bmatrix}, \tag{8}$$

whose dimensions correspond to log-luminance, chroma gain, red–green white balance, blue–green white balance,

detail gain, denoising intent, and shadow lifting. Thus, each atom can be interpreted as a state-conditioned editing rule:

$$\text{near state } c_i \quad \Rightarrow \quad \text{apply edit } e_i. \tag{9}$$

Before training, k-means clustering is performed on pixel states from the training set. The cluster centers initialize the atom centers, and within-cluster standard deviations initialize fixed atom ranges. This data-aware initialization allows different atoms to cover the major luminance, chroma, texture, and noise states in the training data, providing a stable initial division of labor for end-to-end optimization.

3.4. Pixel–Atom Matching and Edit Field

To reduce the computation of high-resolution matching, the state features are first resized to $1/4$ of the input resolution. For location p and atom i, the model computes the normalized state distance:

$$d_{pi} = \frac{1}{2}\sum_{j=1}^{8}\left(\frac{s_{pj} - c_{ij}}{\sigma_{ij} + \varepsilon}\right)^2. \tag{10}$$

The matching weights are produced by a softmax with temperature τ:

$$w_{pi} = \frac{\exp(-d_{pi}/\tau)}{\sum_{k=1}^{K}\exp(-d_{pk}/\tau)}. \tag{11}$$

The edit vector of pixel p is obtained by the weighted combination of all atoms:

$$E_p = \sum_{i=1}^{K} w_{pi}\alpha_i e_i. \tag{12}$$

This gives the low-resolution edit field

$$E^{low} \in \mathbb{R}^{7\times H/4\times W/4}, \tag{13}$$

which is then restored to the input resolution by bilinear interpolation.

The atom weight map $W_i = \{w_{pi}\}$ describes the region where the i-th atom is active. The editing result of any pixel can be decomposed into the contributions of the atoms:

$$C_{pi} = w_{pi}\alpha_i e_i, \qquad E_p = \sum_i C_{pi}. \tag{14}$$

Therefore, the model provides attribution at the state, atom, and spatial levels.

3.5. Differentiable Image Rendering

The renderer $\mathcal{R}$ converts the seven-dimensional edit field into concrete image operations. The input image is first converted to YUV space, and the luminance is decomposed into a low-frequency base and high-frequency detail:

$$Y = Y_b + D. \tag{15}$$

Luminance editing is performed in the log domain:

$$L'_p = \log(Y_{b,p} + \varepsilon) + \lambda_l \tanh(E_{p,0}) + S_p, \tag{16}$$

where S_p is the shadow-lifting term. The enhanced luminance base is

$$Y'_{b,p} = \exp(L'_p) - \varepsilon. \tag{17}$$

The detail gain is defined as

$$\gamma_p^d = 1 + \lambda_d \tanh(E_{p,4}), \tag{18}$$

and the enhanced luminance is

$$Y'_p = Y'_{b,p} + \gamma_p^d D_p. \tag{19}$$

Then, according to the luminance ratio before and after enhancement,

$$r_p = \frac{Y'_p}{Y_p + \varepsilon}, \tag{20}$$

the three RGB channels are scaled synchronously:

$$I_p^l = r_p I_p. \tag{21}$$

This operation preserves the channel ratio of the input pixel and reduces color shifts caused by luminance boosting.

The white-balance parameters form zero-mean log gains:

$$g_p = \begin{bmatrix} b_p^{rg} & 0 & b_p^{bg}\end{bmatrix} - \text{mean}\begin{bmatrix} b_p^{rg} & 0 & b_p^{bg}\end{bmatrix}, \tag{22}$$

$$I_p^{wb} = I_p^l \odot \exp(g_p). \tag{23}$$

The chroma gain is

$$\gamma_p^c = \exp\left(\lambda_c \tanh(E_{p,1})\right), \tag{24}$$

and it is applied to the chroma components:

$$U'_p = \gamma_p^c U_p, \qquad V'_p = \gamma_p^c V_p. \tag{25}$$

Each renderer parameter corresponds to a deterministic image-processing operation. By observing the channels of the edit field, we can directly analyze the brightness lifting, color adjustment, detail recovery, and shadow compensation performed in different regions.

3.6. Spatially Adaptive Denoising

Because noise suppression requires large-scale spatial information, CSEF jointly controls denoising with atom-level denoising intent and a convolutional spatial prior. The controller predicts a single-channel heatmap from intermediate features:

$$Q \in \mathbb{R}^{1\times H/8\times W/8}, \tag{26}$$

and upsamples it to the input resolution. After centering and mean normalization, the heatmap gives a spatial gate:

$$G_p = \text{clip}\left(\frac{\sigma(1.5(Q_p - \bar{Q}))}{\text{mean}_q[\sigma(1.5(Q_q - \bar{Q}))]}, 0.25, 2.0\right). \tag{27}$$

The sixth channel of the edit field gives the atom-level denoising intent, and the final denoising strength is

$$N_p = \mathrm{clip}\left(\sigma(E_{p,5})G_p, 0, 1\right). \tag{28}$$

The model uses guided filtering in YUV space to construct edge-preserving smoothed images:

$$\widetilde{Y} = GF(Y, Y), \qquad \widetilde{U} = GF(U, Y), \qquad \widetilde{V} = GF(V, Y). \tag{29}$$

The raw path and denoised path pass through the same editing renderer, and the final output is

$$\widehat{I}_p = (1 - N_p) I_p^{raw} + N_p I_p^{clean}. \tag{30}$$

Here, N_p can be directly interpreted as the blending ratio of the denoising path in the final result. The atom edit field determines how much denoising is needed for pixels with a certain state, while the spatial heatmap further determines the concrete distribution of denoising operations in the current image.

3.7. Optimization Objective

The model is trained end-to-end with paired low-light and normally exposed images. The total loss consists of a pixel reconstruction loss, a structural similarity loss, and a chroma consistency loss:

$$\mathcal{L} = \lambda_r \mathcal{L}_{char} + \lambda_s \mathcal{L}_{ssim} + \lambda_c \mathcal{L}_{chroma}. \tag{31}$$

The Charbonnier loss constrains pixel-level reconstruction accuracy, the SSIM loss preserves local structure and contrast, and the chroma loss constrains consistency between the prediction and the target in YUV chroma channels. The default setting is

$$\lambda_r = 1.0, \qquad \lambda_s = 0.35, \qquad \lambda_c = 0.25. \tag{32}$$

During training, we use standard low-light–normal-exposure pairs, mixed samples with different low-light and normal-exposure degrees, and normal-exposure identity samples. This data strategy enables the model to learn a continuous editing range that covers strong enhancement, weak enhancement, and identity mapping, improving its adaptability to different input exposure levels.

4. Experiments and Results

4.1. Experimental Settings

We evaluate CSEF on LOL-v2 Real and LOL-v2 Synthetic. Following common settings for low-light image enhancement, we use PSNR, SSIM, and MAE as full-reference reconstruction metrics, and use FPS and parameter count to evaluate model efficiency. Unless otherwise specified, all ablation variants are compared under the same training and testing settings. All runtime results are measured on an NVIDIA GeForce RTX 4060 GPU with input resolution 256×256 and batch size 1. The reported runtime includes model inference and renderer post-processing, and is averaged over all images in the corresponding test set.

The current CSEF backbone consists of an eight-dimensional image state space, 16 data-aware atom centers, a distribution-pooling controller, a seven-dimensional atom edit vector, a denoising heatmap, and a guided-filter rendering path. The model first constructs state features from the low-light image, then uses the controller to generate image-level editing atoms. Each pixel is matched to these atoms in the state space, synthesizing a per-pixel edit field. The renderer interprets the edit field as operations for luminance, color, white balance, detail, and denoising.

4.2. Comparison with Existing Methods

Table 1 reports the quantitative comparison between CSEF and representative low-light enhancement models on LOL-v2 Real and LOL-v2 Synthetic. On LOL-v2 Synthetic, CSEF achieves the best PSNR, SSIM, and MAE among all compared methods, with scores of **22.1841**, **0.8625**, and **0.0728**, respectively. This indicates that the proposed atomic edit field can effectively model synthetic low-light degradation while maintaining a small model size.

On LOL-v2 Real, high-capacity supervised restoration models such as MIRNet [16] and Retinexformer [8] obtain higher pixel-level metrics, but they require more parameters and have lower inference speed. In contrast, CSEF has only about **124K** parameters and reaches **143.46 FPS**. Compared with lightweight methods such as Zero-DCE [4], Zero-DCE++ [17], and SCI [5], CSEF achieves a more stable trade-off between restoration quality and efficiency, showing that state-space atom matching is more expressive than direct curve adjustment.

4.3. Inference-Time Controllability

An important property of CSEF is inference-time controllability. After training, the same checkpoint can be fixed and different enhancement modules can be controlled by adjusting a small number of renderer hyperparameters without retraining. For example, the luminance coefficient can control the overall exposure lifting strength, the white-balance coefficient can control the magnitude of color correction, and detail- or chroma-related coefficients can further adjust the enhancement style.

This controllability comes from the atomic editing mechanism of CSEF. Unlike black-box enhancement models that directly predict the final RGB image, CSEF decomposes enhancement into interpretable editing dimensions, including luminance, color, white balance, detail, and denoising. Therefore, changing the corresponding coefficient at inference time can relatively independently adjust a specific vi-

Table 1. Quantitative comparison on LOL-v2 Real and LOL-v2 Synthetic.

Method	Real PSNR ↑	Real SSIM ↑	Real MAE ↓	Syn. PSNR ↑	Syn. SSIM ↑	Syn. MAE ↓	FPS Real/Syn.	Params
Zero-DCE	18.0588	0.5802	0.1309	17.7564	0.8115	0.1241	112.59 / 195.93	79,416
Zero-DCE++	18.2089	0.5818	0.1248	18.0322	0.8210	0.1162	95.57 / 170.48	10,561
SCI	17.3619	0.5409	0.1460	15.4290	0.7390	0.1564	**214.37 / 311.12**	**258**
IAT	26.4568	0.8432	0.0421	15.3737	0.7043	0.1616	43.97 / 76.05	91,154
MIRNet	28.0960	**0.8892**	0.0349	15.7555	0.7267	0.1436	1.35 / 2.34	31,787,419
Retinexformer	**28.9768**	0.8816	**0.0314**	16.1899	0.7647	0.1348	10.39 / 17.67	1,605,701
CSEF	20.4338	0.7893	0.0893	**22.1841**	**0.8625**	**0.0728**	143.46 / 159.19	124,257

Table 2. Ablation results for major components of CSEF.

Variant	Real PSNR ↑	Real SSIM ↑	Real MAE ↓	Syn. PSNR ↑	Syn. SSIM ↑	Syn. MAE ↓
Baseline, 16 atoms	**20.4338**	**0.7893**	**0.0893**	22.1841	**0.8625**	0.0728
4 atoms	19.4893	0.7517	0.1004	21.7980	0.8540	0.0736
8 atoms	18.9313	0.7641	0.1093	21.2643	0.8529	0.0792
64 atoms	19.4435	0.7617	0.1005	21.3272	0.8571	0.0815
w/o white balance	19.3098	0.7812	0.1004	**22.2189**	0.8591	**0.0726**
w/o noise heatmap	19.6413	0.7663	0.0994	22.1075	0.8602	0.0745
Global edit field	20.1512	0.7572	0.0917	17.3193	0.7747	0.1215
LUT edit table	20.0808	0.7410	0.0897	21.3845	0.8544	0.0789

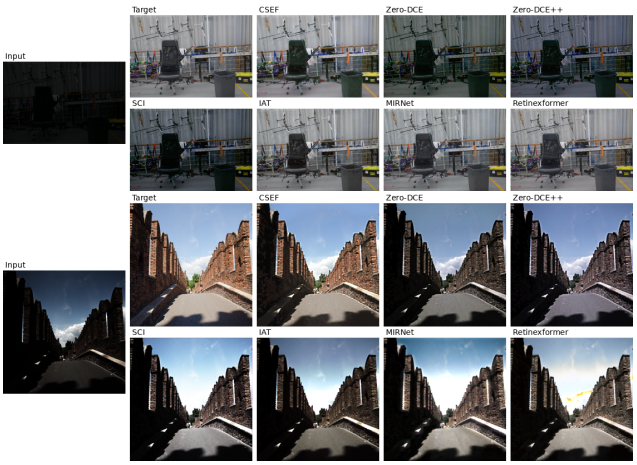


Figure 3. Visual comparison on LOL-v2. CSEF restores clearer luminance and local contrast while keeping the overall tone relatively stable when enhancing dark foreground regions.

sual effect without changing the model parameters.

As shown in Fig. 4, with fixed model parameters, changing only the luminance coefficient produces an output sequence from weak to strong exposure enhancement. Changing only the white-balance coefficient also changes the strength of color correction. This shows that CSEF is not only a lightweight enhancement model but also an adjustable image editing framework. In practice, different scenes, devices, or users may prefer different brightness and color

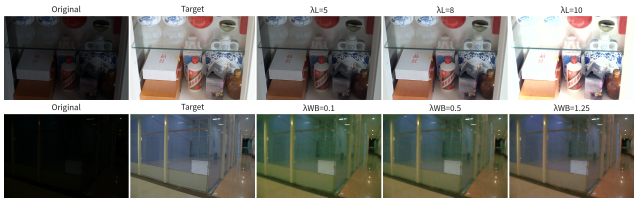


Figure 4. Inference-time controllability of CSEF. The same checkpoint can produce different exposure and white-balance effects by adjusting renderer coefficients.

styles, and CSEF can satisfy such preferences through simple inference-time parameter adjustment.

4.4. Ablation Study and Representation Quality

To verify the effectiveness of each component, we conduct ablation experiments on LOL-v2 Real and LOL-v2 Synthetic. Table 2 reports the results of different variants.

For the number of atoms, the default 16-atom CSEF achieves the most stable overall performance on both datasets. Reducing the atom number to 4 or 8 weakens the expressive ability of the edit field, thereby lowering PSNR and SSIM. Increasing the atom number to 64 does not bring further improvement, indicating that simply enlarging the atom dictionary does not guarantee better enhancement and may instead make atom routing less stable.

For editing components, the white-balance dimension is especially important on LOL-v2 Real. Removing white

Table 3. Representation quality test: indoor/outdoor linear probing.

Feature	Dim.	Train Acc.	Test Acc.
Raw state + average pooling	7	73.25%	73.31%
CSEF controller representation	256	**100.00%**	**92.80%**

balance decreases PSNR from **20.4338** to 19.3098 and increases MAE from **0.0893** to 0.1004. The denoising heatmap also brings stable gains, especially on real low-light images, where removing this branch causes clear performance degradation. This suggests that denoising should not be controlled only by global editing strength but should be spatially adaptive. In addition, although the global-edit variant still performs reasonably on LOL-v2 Real, it degrades markedly on LOL-v2 Synthetic, indicating that pure image-level editing is insufficient for complex local degradation. The LUT edit table also performs worse than the default method, further showing that explicit state-space atom matching is a more stable editing mechanism.

Beyond image enhancement metrics, we test whether the representation learned by the CSEF controller is separable, in order to evaluate its image-level semantic understanding. Specifically, we freeze the trained CSEF controller, train a linear classifier on its distribution-pooling representation, and evaluate it on an indoor/outdoor binary classification task. As shown in Tab. 3, the controller representation reaches **92.80%** test accuracy, while average pooling of the untrained seven-dimensional raw state features reaches only 73.31%. The gap of **19.49** percentage points indicates that the CSEF controller learns more than low-level luminance, color, and texture statistics; it also captures stronger scene-structure information.

4.5. Transfer Experiments

To verify the generality of the atomic editing framework, we further transfer CSEF to two related enhancement tasks: underwater image enhancement and general color enhancement. These experiments examine whether the state-aware atomic editing mechanism can be reused in different enhancement scenarios.

4.5.1. Transfer to Underwater Image Enhancement

Table 4 reports the results on underwater enhancement. Compared with the raw input, CSEF improves PSNR from 17.7596 to **20.1732** and SSIM from 0.7731 to **0.8645**, while reducing MAE and RMSE and improving UCIQE. This indicates that the edit field of CSEF can transfer from low-light enhancement to underwater degradation. UIEC2Net [18], as a model specifically designed for underwater enhancement, achieves higher full-reference metrics. However, CSEF is significantly faster, requiring only **23.54 ms** per image, while UIEC2Net requires 192.87 ms as shown in Tab. 5. Therefore, CSEF can serve as a lightweight cross-domain enhancement framework, and future work can further adjust the color editing dimensions for underwater tasks.

Table 4. Transfer results on underwater image enhancement.

Method	PSNR ↑	SSIM ↑	MAE ↓	RMSE ↓	UCIQE ↑	UIQM ↑
Raw input	17.7596	0.7731	0.1153	0.1506	22.0707	**6.3946**
CSEF edit16	20.1732	0.8645	0.0854	0.1068	25.3892	5.8298
UIEC2Net	**23.1898**	**0.8935**	**0.0619**	**0.0774**	**28.5716**	5.2904

Table 5. Inference-time comparison on underwater enhancement.

Method	ms / image ↓
CSEF v3 edit16	**23.54**
UIEC2Net	192.87

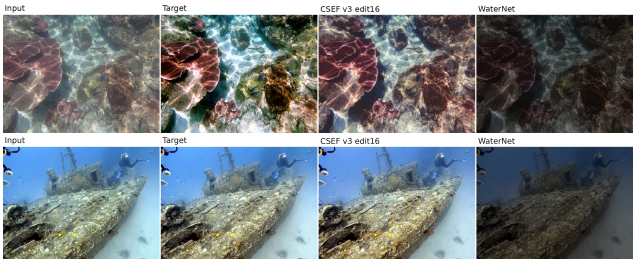


Figure 5. Transfer visualization on underwater image enhancement. CSEF improves visibility and local contrast, while specialized underwater enhancement models can perform stronger color correction.

4.5.2. Transfer to Color Enhancement

We also transfer CSEF to color enhancement using an edit12 configuration that is more suitable for tone and color adjustment. Table 6 shows that CSEF edit12 obtains **22.9149** PSNR, **0.8476** SSIM, and **0.0653** MAE. CSRNet [19], which is designed specifically for color enhancement, achieves higher restoration accuracy and faster inference in this task. Nevertheless, CSEF obtains competitive results without changing the core atom-matching framework. This suggests that the state-space matching and renderer interpretation mechanism of CSEF can be transferred to different image enhancement tasks by adjusting the editing dimensions, without fully redesigning the network.

Table 6. Transfer results on color enhancement.

Method	PSNR ↑	SSIM ↑	MAE ↓	ms/img ↓	FPS ↑
CSEF edit12	22.9149	0.8476	0.0653	7.32	136.57
CSRNet	**23.5749**	**0.8644**	**0.0619**	**5.81**	**171.99**

Overall, the quantitative and qualitative results show that CSEF is not limited to a single low-light dataset. Its atomic editing mechanism can be extended as a general lightweight image enhancement framework.



Figure 6. Transfer visualization on color enhancement. CSEF restores global brightness and scene color while maintaining a relatively natural tone.

5. Conclusion

This paper presented CSEF, a lightweight, interpretable, and controllable framework for low-light image enhancement. Through an eight-dimensional state space, image-conditioned editing atoms, pixel–atom matching, and a differentiable renderer, CSEF decomposes low-light enhancement into explicit operations, including luminance, color, white balance, detail, denoising, and shadow lifting. Experimental results show that CSEF achieves stable enhancement while maintaining a small parameter count and high inference speed. It also provides inference-time controllability, cross-task transferability, and interpretability. A current limitation is that the predefined editing dimensions may be insufficient for severe mixed degradations, and future work will further extend the editing dimensions and task-conditioned rendering mechanisms to improve restoration under more complex scenarios.

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A. Use of AI Tools

During the preparation of this paper, AI tools, including Codex and ChatGPT, were used to assist with related material collection, code writing, and experiment execution.